Squam Lake Water Clarity Model

Validation Report: Compiled Results Across Four Rounds of Testing

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This report compiles every validation pass run against real in-lake Secchi depth readings for Squam Lake and Little Squam Lake, NH, in the order they were performed: the original pooled lake-wide comparison, a Deep Haven-only comparison testing whether averaging across Squam's many bays was diluting a real signal, a correction after discovering the production model had partly trained on Squam-area data, and a final multi-point comparison matching each real reading to a prediction from its own specific monitoring site. All correlations are compared against the national LAGOS-US LANDSAT model's baseline of r = -0.22 (p = 0.26) at these lakes, which is what motivated this project in the first place.

1. Original Pooled Validation Results

The first validation pass compared model predictions at one central coordinate per lake against real Secchi readings averaged across all reporting stations on each lake, for each date. National model baseline at these lakes: r = -0.22, p = 0.26 (no real relationship).

Exact date match

Comparison	n	Pearson r	p-value	R2	MAE (m)	RMSE (m)
Both lakes pooled	23	0.061	0.7833	0.004	1.816	2.200
Squam Lake only	14	-0.164	0.5747	0.027	1.902	2.394
Little Squam Lake only	9	0.349	0.3574	0.122	1.682	1.858

+/-3-day tolerance match

Comparison	n	Pearson r	p-value	R2	MAE (m)	RMSE (m)
Both lakes pooled	97	0.212	0.0373	0.045	1.909	2.253
Squam Lake only	52	0.175	0.2155	0.031	1.765	2.155
Little Squam Lake only	45	0.271	0.0720	0.073	2.076	2.361

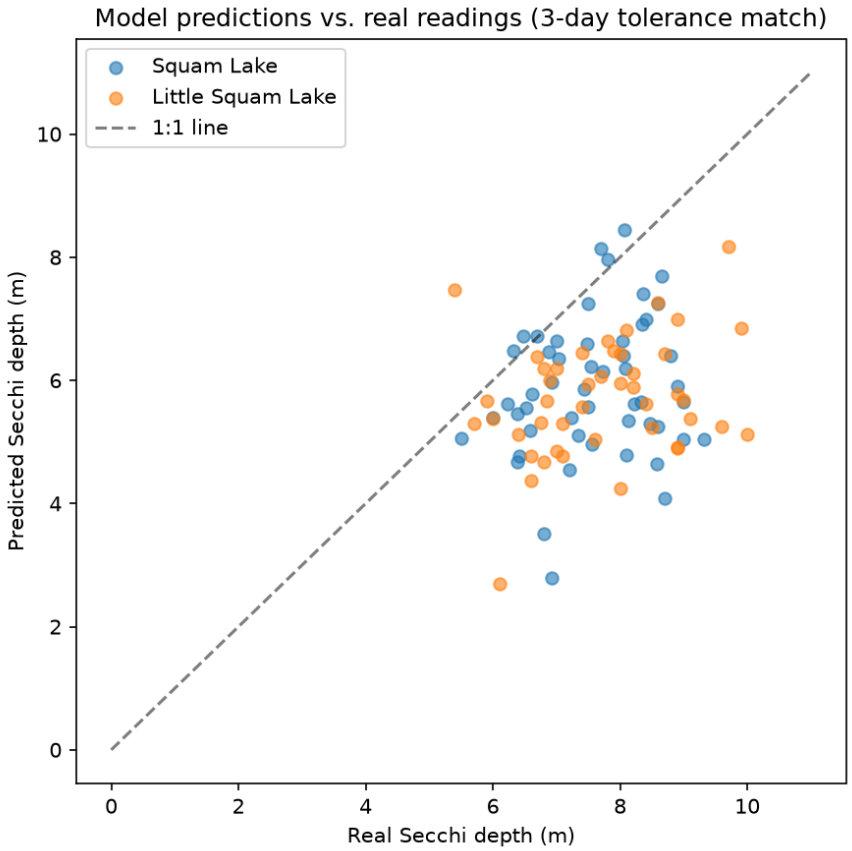


Figure 1. Predicted vs. real Secchi depth, both lakes pooled at their central coordinates, +/-3-day tolerance match.

2. Deep Haven-Only Results

Squam Lake is a large, multi-basin lake, and the real-readings file showed many stations sampled on the same date -- averaging across all of them could dilute a real, localized signal. This pass restricted Squam's comparison to a single consistent monitoring site, Deep Haven, while leaving Little Squam's comparison unchanged (lake-wide average, only one site available for that lake).

Exact date match

Comparison	n	Pearson r	p-value	R2	MAE (m)	RMSE (m)
All (Squam/Deep Haven + Little Squam)	21	0.202	0.3791	0.041	2.177	2.519
Squam Lake (Deep Haven only)	12	0.089	0.7832	0.008	2.548	2.918
Little Squam Lake (unchanged)	9	0.349	0.3574	0.122	1.682	1.858

+/-3-day tolerance match

Comparison	n	Pearson r	p-value	R2	MAE (m)	RMSE (m)
All, tolerance	86	0.312	0.0035	0.097	2.377	2.692
Squam Lake (Deep Haven only)	41	0.323	0.0392	0.105	2.708	3.014
Little Squam Lake (unchanged)	45	0.271	0.0720	0.073	2.076	2.361

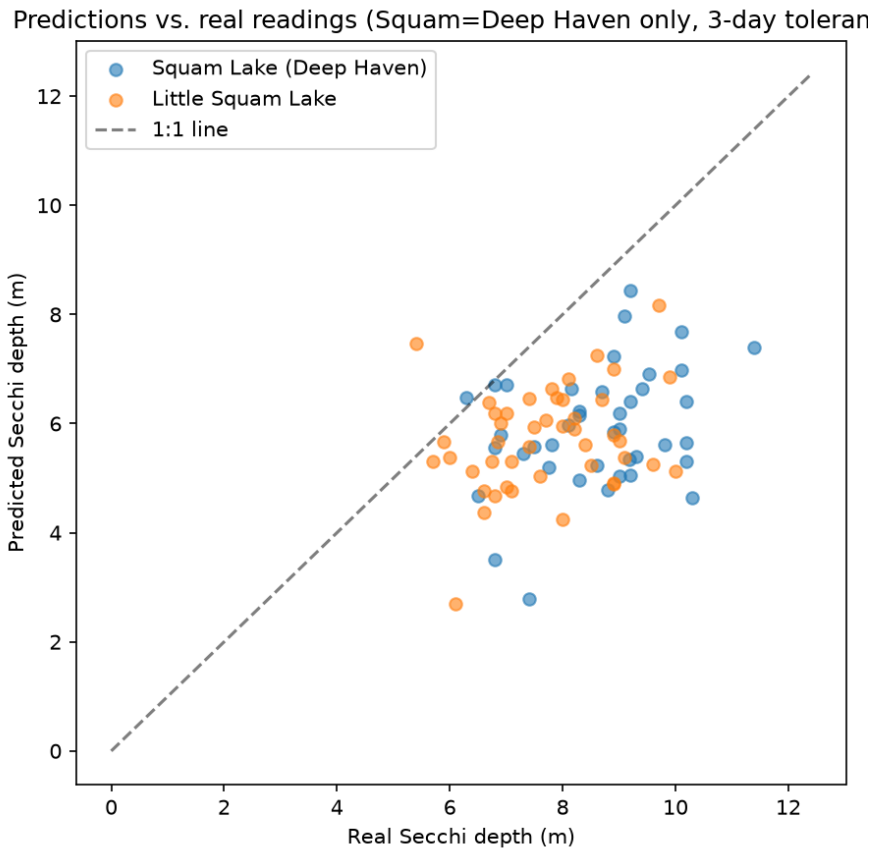


Figure 2. Predicted vs. real Secchi depth, Squam restricted to the Deep Haven site, +/-3-day tolerance match. Correlation improved for Squam under the looser tolerance match, but absolute error (MAE) got worse, not better -- a partial, not a confirming, result.

3. Data Leakage Correction

A review of the training data found that 107 rows, drawn from 16 monitoring sites, were genuine Squam Lake and Little Squam Lake stations (the same NH DES monitoring network used in the real validation data, e.g. "SQUAM LAKE-DEEP HAVEN", "SQUAM LAKE-LOON REEF") included in the pool of ~19,500 rows used to train the production model. The spatial cross-validation figure reported during model development (r = 0.512, testing on entirely unseen lakes) remained valid, since it used group-based cross-validation that held out whole lakes in rotation. But the single final model used to generate the actual Squam/Little Squam predictions had been trained on all rows, including these 107 -- meaning the validation comparisons above were not fully independent tests for the overlapping records.

The model was retrained excluding every Squam-area site entirely (19,403 rows, 1,383 sites remaining). Spatial cross-validation on this reduced pool was nearly identical to before (r = 0.507 vs. 0.512), indicating the excluded rows had negligible influence on the model overall -- but the corrected model is the only one that has genuinely never seen Squam or Little Squam data, so it is the valid one for the remaining comparisons in this report.

Corrected: Exact date match

Comparison	n	Pearson r	p-value	R2	MAE (m)	RMSE (m)
Both lakes pooled (lake-wide)	23	0.112	0.6100	0.013	1.799	2.177
Squam Lake only (lake-wide)	14	-0.123	0.6758	0.015	1.895	2.388
Little Squam Lake only	9	0.432	0.2453	0.187	1.650	1.801
Squam Lake (Deep Haven only)	12	0.130	0.6879	0.017	2.485	2.872

Corrected: +/-3-day tolerance match

Comparison	n	Pearson r	p-value	R2	MAE (m)	RMSE (m)
Both lakes pooled (lake-wide)	97	0.235	0.0207	0.055	1.907	2.243
Squam Lake only (lake-wide)	52	0.194	0.1678	0.038	1.785	2.169
Little Squam Lake only	45	0.302	0.0438	0.091	2.048	2.326
Squam Lake (Deep Haven only)	41	0.335	0.0323	0.112	2.707	3.017

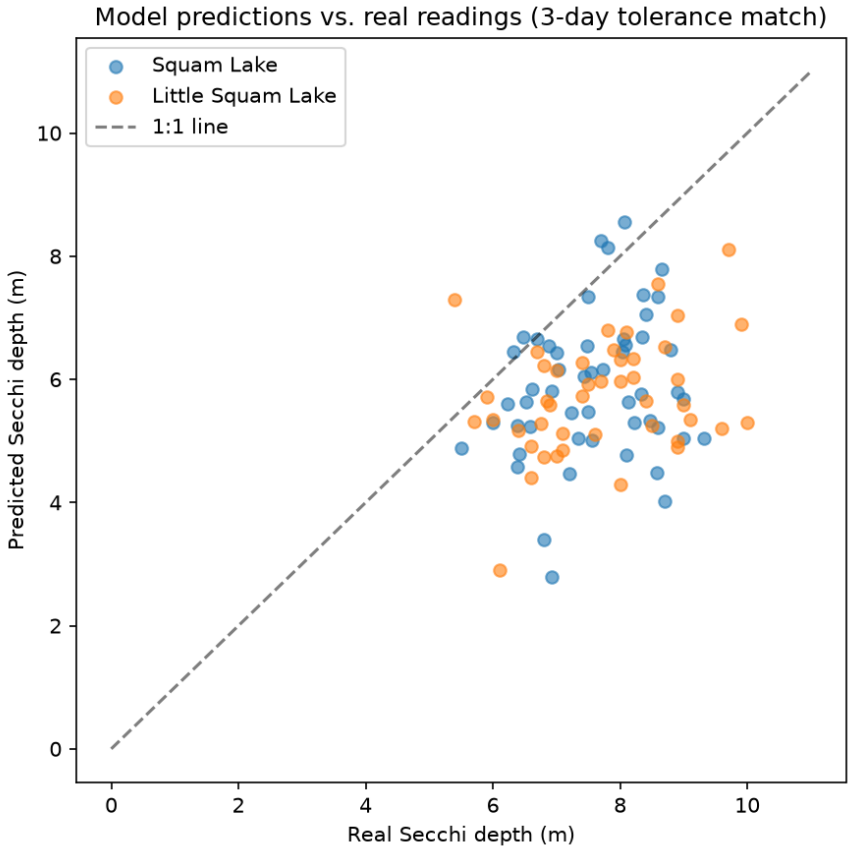


Figure 3a. Corrected model, lake-wide comparison, +/-3-day tolerance match.

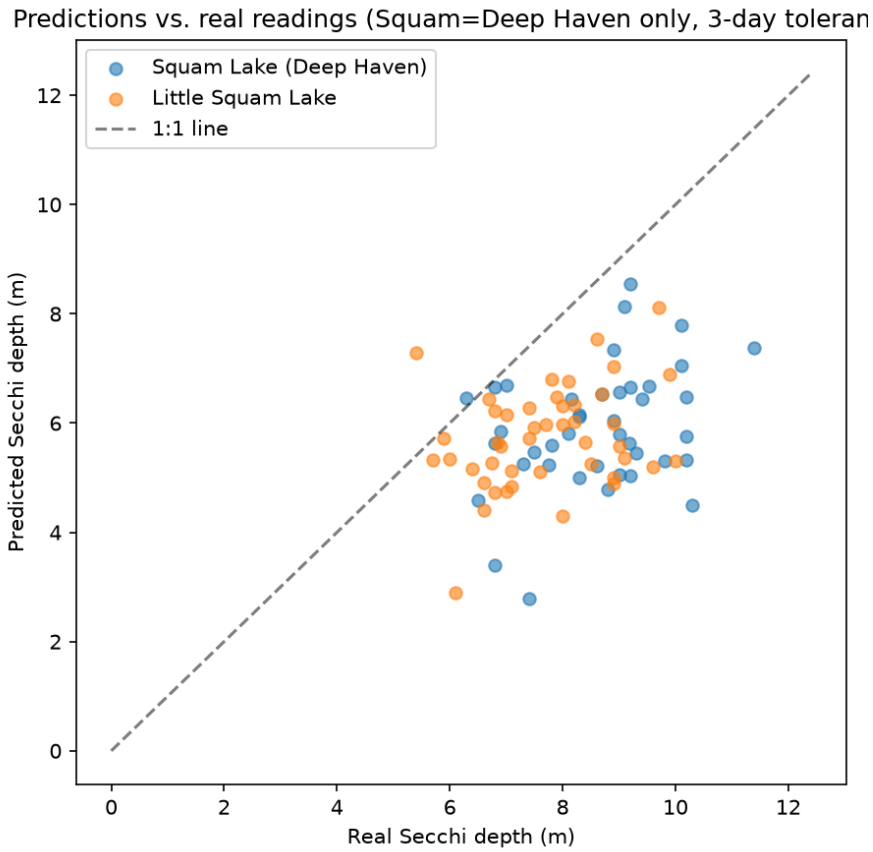


Figure 3b. Corrected model, Squam restricted to Deep Haven, +/-3-day tolerance match.

4. Final Multi-Point Spatial Results

The final validation pass used real GPS coordinates for 13 named monitoring sites across Squam and Little Squam, extracting a separate Landsat-based prediction at each site (instead of one central point per lake), and matching each real reading to the prediction from its own site using the site labels already present in the real-readings data. Predictions used the corrected, Squam-excluded model. Eight real-data site names had no matching coordinate provided and were excluded rather than guessed at: 1 East, 13 Bean Cove, 14 Strutevant Bay, 14A Inner Sturtevant Bay, 16A Inner Dog Cove, 1B, 20 Bear Cove, 9B Outer Squaw Cove.

Pooled across all 13 matched sites

Match type	n	Pearson r	p-value	R2	MAE (m)	RMSE (m)
Exact date	135	0.191	0.0268	0.036	1.898	2.318
+/-3-day tolerance	445	0.264	<0.0001	0.070	1.908	2.292

Per-site breakdown (+/-3-day tolerance match)

Site	n	Pearson r	p-value	R2	MAE (m)
5 Livermore Cove	37	0.463	0.0039	0.215	1.623
2 Cotton Cove	29	0.344	0.0676	0.118	1.123
8 Rattlesnake Cove	32	0.317	0.0774	0.100	1.073
9A Inner Squaw Cove	25	0.266	0.1979	0.071	1.082
12 Moultonboro Bay	38	0.224	0.1757	0.050	2.159
10 Sandwich Bay	39	0.204	0.2125	0.042	2.699
Deep Haven	41	0.186	0.2451	0.034	2.773
1 West (Little Squam)	52	0.165	0.2423	0.027	2.029
16 Dog Cove	33	0.143	0.4266	0.021	1.259
Loon Reef	36	0.146	0.3966	0.021	2.885
18 Piper Cove	44	0.119	0.4420	0.014	1.447
14 Sturtevant Bay	37	0.101	0.5503	0.010	2.093
11 Kent Island	2	—	too few pairs	—	—

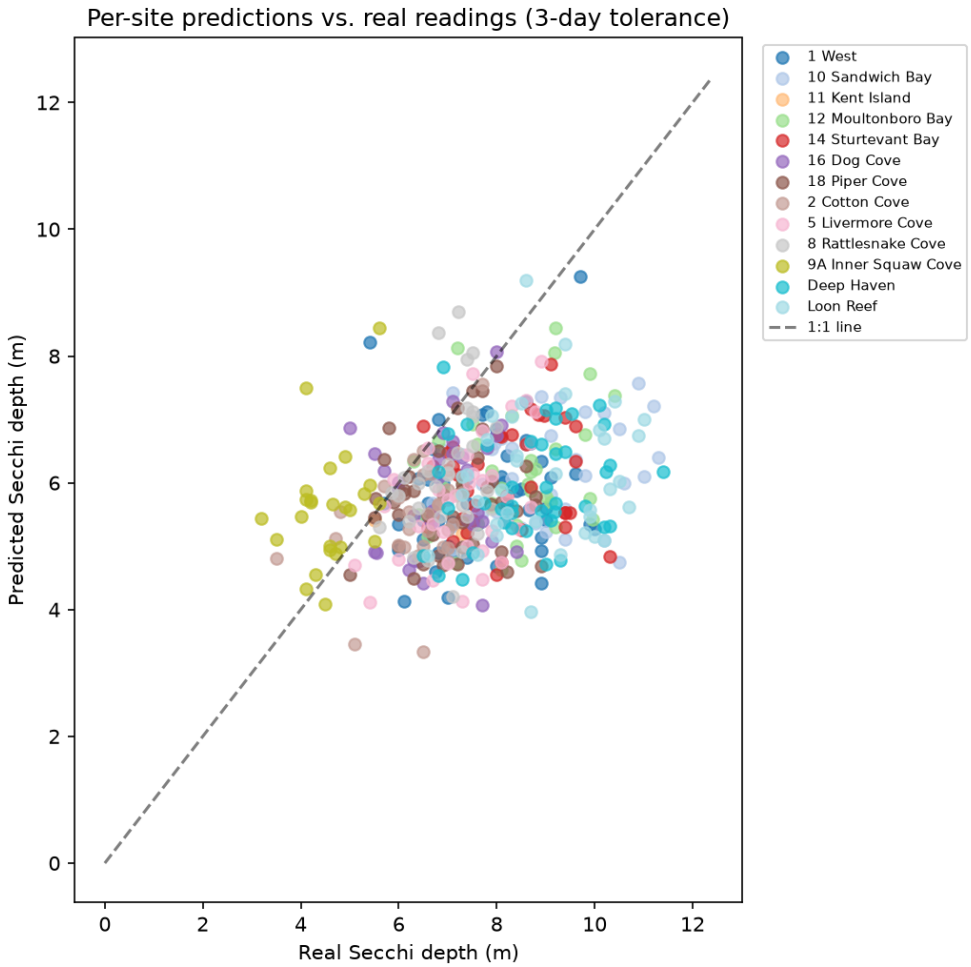


Figure 4. Per-site predictions vs. real readings, all 13 matched sites, +/-3-day tolerance match. Coves and nearshore sites (Livermore, Rattlesnake, Cotton) show the strongest, lowest-error relationships; the two open-water sites (Deep Haven, Loon Reef) show the weakest correlation and the largest errors.

Code and Data

All code, prediction outputs, and validation reports for this project are available at:

https://github.com/fgajarsa-sudo/squam-water-clarity-model

The repository contains the full processing pipeline (12 numbered Python scripts covering data collection from the EPA Water Quality Portal, Landsat matching via Google Earth Engine, feature engineering, model training with spatial hold-out cross-validation, prediction generation, and all four rounds of validation described in this report), the final prediction CSVs/Excel files for Squam and Little Squam (1984-present), and every validation report and scatter plot referenced above. Raw and intermediate data files are intentionally excluded (large, reproducible from the public EPA and Earth Engine APIs, and some raw EPA fields contained incidental volunteer field notes unrelated to this project that were stripped before publication).